

## AI Use Declaration Form

**Course:** Introduction to Artificial Intelligence

**Lab Title:** Lab 1 Part A — Uninformed Search Algorithms

**Student Name:** \_\_\_\_\_ Janet Dankwah

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**Student ID:** \_\_\_\_\_ 23112028 \_\_\_\_\_

**GitHub Repository Link:** \_\_\_\_\_

**Date Submitted:** \_\_\_\_\_ 17/06/26 \_\_\_\_\_

### 1. AI Use Summary

| Question                                            | Student Response |
|-----------------------------------------------------|------------------|
| Did you use any AI tool for this lab?               | <b>yes</b>       |
| Estimated percentage of the work influenced by AI   | 15 %             |
| Did you attach evidence of AI use where applicable? | <b>yes</b>       |

### 2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

| Name of AI Tool Used | Purpose for Using the Tool | Prompt or Instruction Given to the Tool                                                                            | Part(s) of the Work Influenced by the Tool | How I Verified, Edited, or Improved the AI Output         |
|----------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-----------------------------------------------------------|
| Chat GPT             | Debugging                  | 'I followed the pseudocode in the lab, but it gives me errors when I run it. Can you help me identify the error??' | At the return node or state section        | I was returning node instead of the searchResult function |
|                      |                            |                                                                                                                    |                                            |                                                           |
|                      |                            |                                                                                                                    |                                            |                                                           |

### 3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

| Evidence Type | Attached? (Yes, | File Name |
|---------------|-----------------|-----------|
|---------------|-----------------|-----------|

|                                       |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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|                                       | No,<br>N/A) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| AI-generated draft                    |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Prompt history                        |             | <div> <div>summary</div> <p>For Lab 1A, a raw summary of the prompts you sent would be:</p> <ul style="list-style-type: none"> <li>• Setting up the AI lab environment (Python, Jupyter Notebook, VS Code, GitHub)</li> <li>• Asking which notebook/file to work in.</li> <li>• Understanding the drone pathfinding problem formulation.</li> <li>• Explaining BFS, DFS, DLS, and IDS algorithms.</li> <li>• Understanding nodes, states, frontiers, and explored/reached sets.</li> <li>• Translating pseudocode into Python.</li> <li>• Debugging implementation errors after following the pseudocode.</li> <li>• Fixing issues with queues, stacks, and frontier management.</li> <li>• Understanding path-cycle checking and repeated-state checking.</li> <li>• Comparing BFS, DFS, DLS, and IDS behavior.</li> <li>• Troubleshooting code that would not run or produced incorrect output.</li> <li>• Clarifying object-oriented programming requirements for the search algorithms.</li> <li>• Verifying that implementations matched the lab specifications.</li> <li>• Understanding search results such as path cost, nodes expanded, and solution p</li> <li>• Asking for help interpreting assignment requirements and expected outputs.</li> </ul> <p>That's essentially the progression of our Lab 1A conversations: <b>environment setup</b> → <b>understanding</b> → <b>pseudocode translation</b> → <b>debugging</b> → <b>result interpretation</b>.</p> </div> |
| Screenshot of AI interaction          |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Exported AI conversation              |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| AI-generated code snippet             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Revised version showing student input |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

#### 4. Student Declaration

I declare that:

| Declaration Statement                                                                     | Response<br>(Yes/No) |
|-------------------------------------------------------------------------------------------|----------------------|
| The submitted work is my own work.                                                        | Yes                  |
| Any use of AI tools has been clearly declared in this form.                               | Yes                  |
| The prompts, instructions, and AI-generated outputs have been disclosed where applicable. | Yes                  |
| I reviewed, tested, edited, and improved any AI-generated content before submission.      | Yes                  |
| My AI usage does not exceed 25% of the entire work.                                       | Yes                  |
| I understand that undeclared or excessive AI use may be treated as academic misconduct.   | Yes                  |

**Student Signature:** \_\_\_\_\_ Janet Dankwah Nortey \_\_\_\_\_

**Date:** \_\_\_\_\_ 17/06/26 \_\_\_\_\_